# Elhadji Diaraff DIAGNE

**Senior Data Scientist**

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## Professional Summary

Data Scientist with over 6 years of experience in data science, machine learning, fraud detection, Business Intelligence (BI), and large-scale data analysis across Cloud environments such as JupyterHub, SAS Viya, and Databricks. Proven expertise in designing, developing, and deploying high-value use cases to improve operational performance, drive strategic decision-making, and support organizational digital transformation.

## Technical Skills

- **Languages:** Python, PySpark, SQL, SAS (Studio & Analytics), R, Bash
- **Machine Learning & Deep Learning:** Scikit-Learn, TensorFlow, PyTorch, Keras, XGBoost, Hugging Face
- **Generative AI & LLM:** OpenAI API, LangChain, RAG, Prompt Engineering
- **Data Wrangling & Scraping:** Pandas, NumPy, Scrapy, BeautifulSoup
- **Data Engineering & ETL/ELT:** Apache Airflow, Google Dataflow, dbt, Apache Spark
- **Data Warehousing & Cloud:** Google BigQuery, Databricks, Cloudera, Snowflake, GCP
- **Visualization & Web:** Power BI, SAS Visual Analytics, D3.js, Chart.js, Streamlit, FastAPI
- **MLOps, DevOps & Tools:** Git, Docker, CI/CD, MLflow

## Work Experience

**Data Scientist** Nov. 2020 – Present

*Orange Mali* *Bamako, Mali*

- **Fraud & Revenue Assurance Use Cases:** Designed and deployed predictive analytics solutions and executive dashboards on massive telecom data volumes (**SAS Viya, Python, PySpark, SQL, Power BI**), optimizing fraud detection and revenue preservation.
- **Web & AI Platform Development (FastAPI, PySpark, Hive):** Engineered a multi-module enterprise web application connected to Hive via PySpark with role-based access control:
  - **Activity Tracking:** Dynamic forms and interactive KPIs to track project deployments and automation levels.
  - **Fast Client Profiling:** Instant customer behavior analysis module based on Voice, SMS, and Data usage patterns.
  - **Fraud Monitoring:** Integrated real-time decision dashboards dedicated to network anomaly surveillance.
  - **Scheduling & MLOps:** Script and ML model execution pipeline interface (.py, .ipynb, .sh) providing full user autonomy.
  - **AI Assistant / GenAI (RAG):** Deployed a RAG solution over aggregated data, enabling natural language querying of business KPIs to accelerate decision-making.
- **Social Media & Web Scraping:** Built an automated web extraction pipeline (**Facebook-Scraper, Streamlit**) for behavioral analysis and reverse-engineering of fraudsters' operational methods on mobile Data services.
- **Predictive Machine Learning Modeling:** Trained, evaluated, and deployed an **XGBoost** model for proactive Simbox fraud detection on the telecom network.

**Data Analyst** Aug. 2020 – Nov. 2020

*Centre de services scolaire de Montréal (CSDM)* *Montreal, Canada*

- **Academic Success KPI Tracking:** Processed, analyzed, and modeled student learning data to provide school advisors with actionable insights and high-quality reporting dashboards.
- **Data Collection & Digitalization:** Built web applications for online data collection to improve data centralization and accuracy.

**AI Developer** Mar. 2020 – Jul. 2020

*Innovation Maritime* *Rimouski, Canada*

- **Data Engineering & ETL:** Built an ETL data pipeline for maritime data hosted on SQL Server using R's RODBC framework for sea level prediction models across Vieux-Québec, Saint-François, and Saint-Joseph stations.
- **Predictive Time-Series Modeling:** Evaluated time-series architectures (ARIMA, Prophet, LSTM) and deployed an ARIMA model executing sea level predictions every 3 minutes.
- **Automation & Monitoring:** Automated export flows to OneDrive and developed a dynamic real-time monitoring system (Dygraphs / R) to track discrepancies between predicted and observed values.

**Research & Teaching Assistant** May 2018 – Sep. 2019

*UQTR – Department of Mathematics and Computer Science* *Trois-Rivières, Canada*

- **Applied Research:** Conducted comparative research between traditional statistical methods (Discriminant Analysis, Regression) and modern algorithms (Machine Learning, Multilayer Perceptron) for supervised classification.
- **Teaching & Academic Support:** Mentored and assisted undergraduate students in probability and statistics courses.

**Statistician, Regional Lead (RGE)** Feb. 2016 – Jan. 2018

*National Agency for Statistics and Demography (ANSD)* *Dakar, Senegal*

- **Field Operations Supervision:** Trained and managed 15 field teams and regional supervisors across Diourbel and Louga regions to ensure high data quality during the general business census.
- **Data Processing & Reporting:** Cleansed, managed, and analyzed collected census data, producing analytical reports for decision-makers.

## Projects & Achievements

**AI Assistant & Tabular GenAI** | *Python, LLM, RAG, LangChain*

- **Wiito Team Member (3 Data Scientists):** Developed and deployed a LLM / RAG solution designed to automate the analysis and natural language querying of complex tabular datasets.

**FASTProfiler Platform** | *FastAPI, SAS Viya, SQL, Geolocation*

- **Fast Customer Profiling:** Designed a dynamic analytics platform providing summary insights into customer behavior and identity attributes (Voice, SMS, Data usage, plan subscriptions, geolocation) to rapidly identify fraud profiles.

**Customer Churn Modeling** | *PySpark, Azure Databricks, Big Data*

- **Attrition Analysis:** Deployed a Machine Learning pipeline on large-scale telecom datasets to predict, analyze, and mitigate customer churn rates.

**Hydrographic Time-Series Forecasting** | *Python, XGBoost, Machine Learning*

- **Water Level Prediction:** Developed and calibrated an XGBoost Machine Learning model for predictive sea level forecasting across monitoring stations in Quebec.

**Telecom Fraud Use Cases Industrialization** | *SAS Viya, PySpark, SQL, Python*

- **Fraud Detection & Revenue Assurance:** Deployed over 30 telecom fraud detection and prediction use cases across Orange Mali's network.

**Threat Intelligence & Social Media Scraping** | *Python, Web Scraping, NLP*

- **Modus Operandi Analysis:** Engineered automated web scrapers (Google, Facebook, YouTube) to gather public sentiment and analyze emerging mobile Data fraud techniques.

## Education

**Master's Degree in Applied Mathematics and Computer Science** Jan. 2018 – Sep. 2019

*Université du Québec à Trois-Rivières (UQTR)* *Quebec, Canada*

- **Research Thesis:** Discriminant Analysis and Multilayer Perceptron: formal links and applications. [Thesis Sep. 2019]

**Bachelor's Degree in Applied Statistics and Computer Science** Nov. 2016 – Aug. 2017

*INEFSAGEP* *Dakar, Senegal*

**Associate Degree in Statistics (Technicien Supérieur)** Oct. 2013 – Jul. 2015

*National School for Statistics and Economic Analysis (ENSAE)* *Dakar, Senegal*

## Certifications

- **Data Science in Azure Databricks** *[In Progress]*
- **Machine Learning with PySpark in Azure Databricks** *Coursera [2026]*
- **Python for Data Science** *IBM [2020]*

## Languages & Interests

- **Languages:** French (Fluent / Native), English (Intermediate), Arabic (Intermediate)
- **Interests:** Soccer, Reading, Traveling